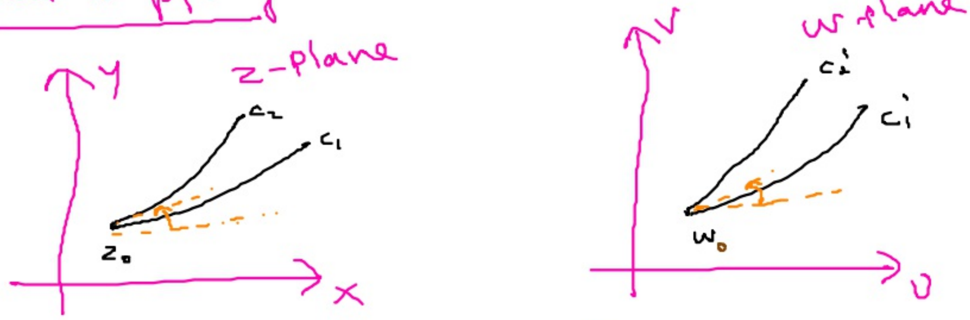
Conformal mapping



Under the transformation $w = f(z)$, a point z_0 and any two curves c_1 and c_2 passing thro' z_0 in the z-plane will be mapped onto a point w_0 and two curves c_1' and c_2' in the w-plane.

If the angle b/w c_1 and c_2 at z_0 is the same as the angle b/w c_1' and c_2' at w_0, both in magnitude and sense. Then such a transformation $w = f(z)$ is said to be conformal at z_0.

Note: A transformation $w = f(z)$ is conformal at a point z_0 if $f(z)$ is analytic at z_0 and $f'(z_0) \neq 0$.

Special conformal transformations

* $w = z^2$ * $w = e^z$

Square transformation

The Square transformation is given by

$$w = z^2 \quad \text{--- (1)}$$

put $w = u+iv$, $z = x+iy$ in (1)

$$u+iv = (x+iy)^2$$

(ie) $\quad = x^2-y^2+2ixy$

$\Rightarrow u = x^2-y^2$ — (2)
$\quad v = 2xy$ — (3)
} eqn. of transformation.

* Consider a St. line $\parallel^{el}$ to X-axis in z-plane, $y = a$, where a is a constant,

$$u = x^2 - a^2 \quad \& \quad v = 2xa$$

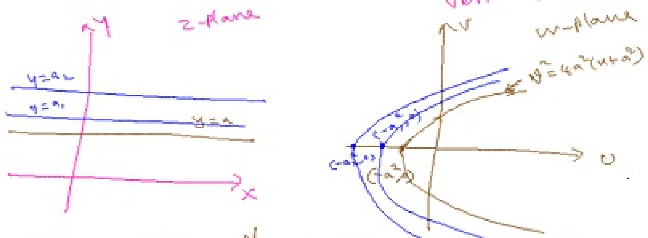
$$\Rightarrow x = \frac{v}{2a}$$

$\therefore u = x^2 - a^2$ becomes

$$u = \frac{v^2}{4a^2} - a^2$$

(ie) $\dfrac{v^2}{4a^2} = u + a^2$

(ie) $v^2 = 4a^2(u+a^2)$ → a parabola, open to the right, with focus $(0,0)$ and vertex $(-a^2, 0)$



Here, a st. line $\parallel^{el}$ to x-axis in z-plane is mapped onto a parabola open to the right in w-plane, as shown in fig.

In $y = a$, if a is an arbitrary constant, then $y = a$ represents system of St. lines, $\parallel^{el}$ to x-axis, in z-plane.

$\therefore$ The sym of St. Lines $\parallel^{el}$ to x-axis in z-plane are mapped onto a family of Confocal parabola in w-plane, as shown in diag.

Parabolae, having common focus.

* Consider a St. line, $\parallel^{el}$ to Y-axis in z-plane, Say $x = b$, where b is a constant.

Using $x=b$ in ② & ①, we have

$$u = b^2 - y^2 \quad \text{and} \quad v = 2by$$

$$\Rightarrow y = \frac{v}{2b}$$

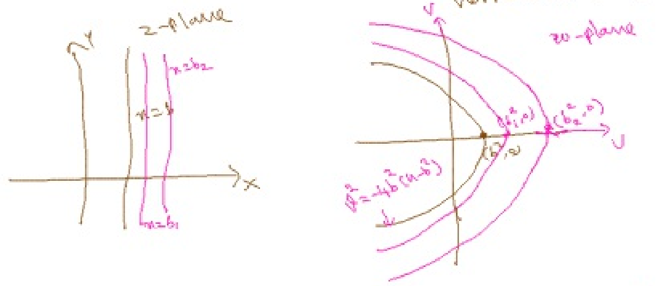
$$\therefore \; u = b^2 - \left(\frac{v^2}{4b^2}\right)$$

$$\text{(i.e)} \; \frac{v^2}{4b^2} = b^2 - u$$

$$= -(u - b^2)$$

$$\text{(i.e)} \; v^2 = -4b^2(u - b^2)$$

a parabola, open to the left, having focus $(0,0)$ and vertex at $(b^2,0)$



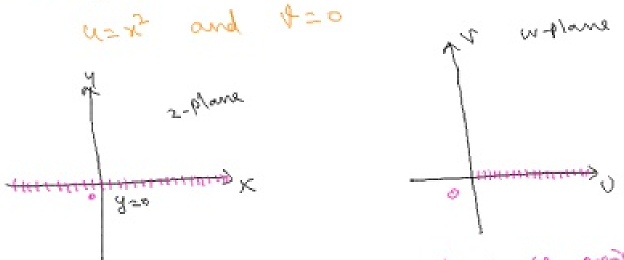
→ A St. line $\parallel^{el}$ to y-axis in z-plane is mapped onto a parabola, open to the left, as shown in fig.

→ When b is an arbitrary constant, $x=b$ represents a system of St. line $\parallel^{el}$ to y-axis in z-plane, for which the image is a sym of confocal parabola in w-plane, as shown in fig.

* $y=0$ in z-plane,

using $y=0$ in ② & ③, we have

$$u = x^2 \quad \text{and} \quad v = 0$$

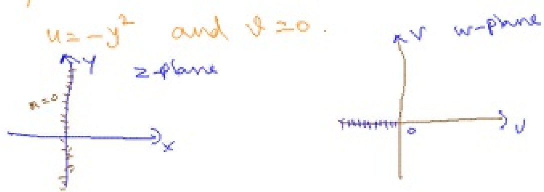


Thus $y=0$ in z-plane is mapped onto the positive part of the u-axis in w-plane

* $x=0$ in z-plane

Using $x=0$ in ② & ③, we have

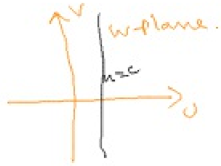
$$u = -y^2 \quad \text{and} \quad v = 0.$$

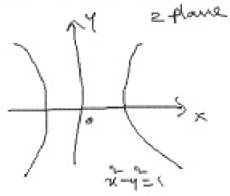


Thus, $x=0$ in z-plane is mapped onto the (−ve) part of the u-axis in w-plane

* $u = c$, represents a family of st. lines $\|^{el}$ to v-axis in w-plane.
  ↳ arbitrary const.

② ⇒ $x^2 - y^2 = c$, represents a family of rectangular hyperbolas in z-plane.

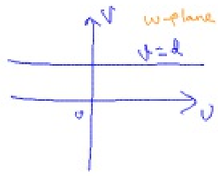


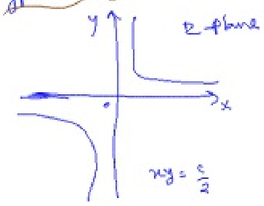


* $v = d$, represents a st. line $\|^{el}$ to u-axis in w-plane.
  ↳ an arbitrary const.

Using ③, $2xy = d$

$\Rightarrow xy = \dfrac{d}{2}$, a family of rectangular hyperbolas in z-plane.





* Conformality

$w = z^2$ (i.e) $f(z) = z^2$

$\Rightarrow f'(z) = 2z$, $f'(z) = 0$, for $z = 0$

$\Rightarrow f(z) = z^2$ is conformal everywhere, except at $z = 0$

Exponential transformation $(w = e^z)$ — (1)

put $z = x+iy$, $w = Re^{i\phi}$

$\therefore (1) \Rightarrow Re^{i\phi} = e^{x+iy}$

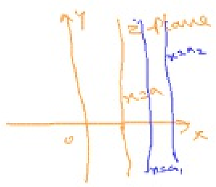
$$\left.\begin{aligned} \Rightarrow\ & R = e^x \\ \& \ & \phi = y \end{aligned}\right\} \text{equn. of transformation.}$$

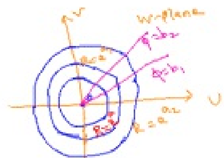
* consider a sym of st-lines, $\parallel^{el}$ to y-axis in z-plane, say $x = a$ ↳ arbitrary constant.

$\therefore R = e^x$ becomes

$\underset{\sim}{R} = \underset{\text{constant}}{\underset{\sim}{e^a}}$ ⟶ represents a circle with $c(0,0)$ and $r = e^a$, in w-plane as shown below.

$|w| = \sqrt{u^2+v^2}$



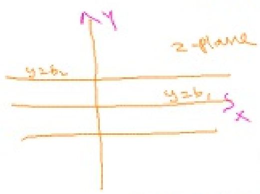


Thus, the image of family of st-lines, $\parallel^{el}$ to y-axis in z-plane is given by family of concentric circles in the w-plane, as shown in diagram.

* consider a family of st-line, $\parallel^{el}$ to x-axis in z-plane, say $y = b$ ↳ arbitrary constt.

$\therefore \phi = y$ becomes

$\phi = b$ ↳ may be $b_1, b_2, \ldots$



Thus, image of family of st-lines, $\parallel^{el}$ to the x-axis, in z-plane is given by family of concurrent lines (lines intersecting at a point) through the origin in the w-plane.

* $x=0$ in z-plane

$R = e^0$

(ie) $R=1 \longrightarrow$ represents $u^2+v^2=1$

↳ unit circle with $C(0,0)$ in w-plane.

* Image of $x<1$ in z-plane is given by

$u^2+v^2<1$ in w-plane.

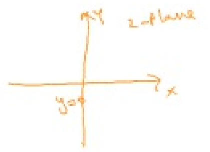
↳ an interior of the circle

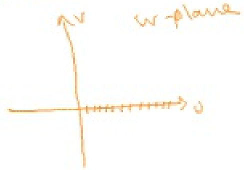
* Image of $x>1$ in z-plane is given by

$u^2+v^2>1$ in w-plane, an exterior portion of the unit circle $u^2+v^2=1$.

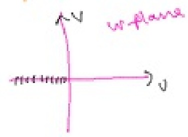
* $y=0$ in z-plane gives

$\phi=0$ in w-plane

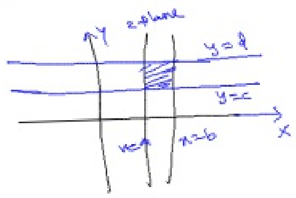


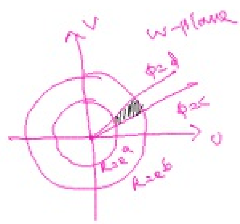


* $y=\pi$ in z-plane gives $\phi=\pi$ in w-plane



* $a \le x \le b$; $c \le y \le d$ in z-plane.





$a \le x \le b \implies e^a \le R \le e^b$

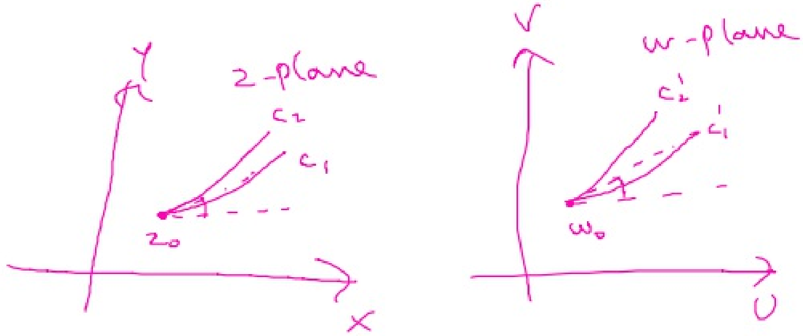
& $c \le y \le d \implies c \le \phi \le d$

* Conformality $w=e^z$

(ie) $f(z)=e^z \implies f'(z)=e^z$

And e^z does not vanish for any finite value of z,

$\implies e^z$ is conformal everywhere



$f'(z_0) \neq 0$

$w = z^2$

$f'(z) = 2z = 0$

$z = 0$ except

$z \neq 0$

$u(x,y) = \dfrac{\sin 2x}{\cosh 2y - \cos 2x}$ $\quad f(z) =$

$$u_x = \frac{(\cosh 2y - \cos 2x)\cdot 2\cos 2x - \sin 2x\,(2\sin 2x)}{(\cosh 2y - \cos 2x)^2}$$

$x \to z$, $y \to 0$

$$\frac{(1 - \cos 2z)\, 2\cos 2z - 2\sin^2 2z}{(1 - \cos 2z)^2}$$

$$f'(z) = u_x \quad \frac{2\cos 2z - 2}{(1 - \cos 2z)^2} = \frac{-2(1 - \cos 2z)}{(1 - \cos 2z)^2} = \frac{-2}{1 - \cos 2z}$$

$$f'(z) = \frac{-2}{\underbrace{1 - \cos 2z}_{2\sin^2 z}}$$

$$u_y = \frac{-\sin 2x\,(2\sinh 2y)}{(\cosh 2y - \cos 2x)^2} = 0$$

$\sin^2 x = \dfrac{1 - \cos 2x}{2}$

$f'(z) = -\operatorname{cosec}^2 z$

$f(z) = \cot z$